U5 Summary

“This pdf covers <u>most</u> of the details.”

Have fun with this gorgeous syllabus 😍😇

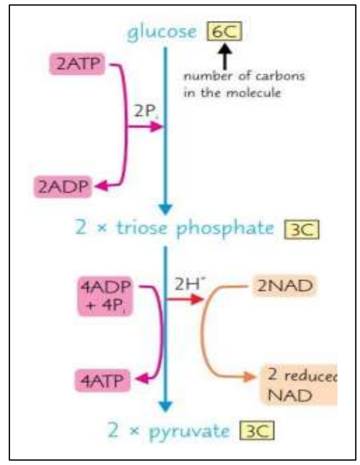
Made by: Nour

TOPIC 7

Respiration

Aerobic respiration:

1- Glycolysis: "In cytoplasm."



- Phosphorylation of glucose using 2 ATP.
    - ➢ Giving us 2 trios phosphate.
- 2 trios phosphate are oxidized giving us 2x pyruvate.
- Then NAD will be reduced to give us 2x NADH and 2x ATP.

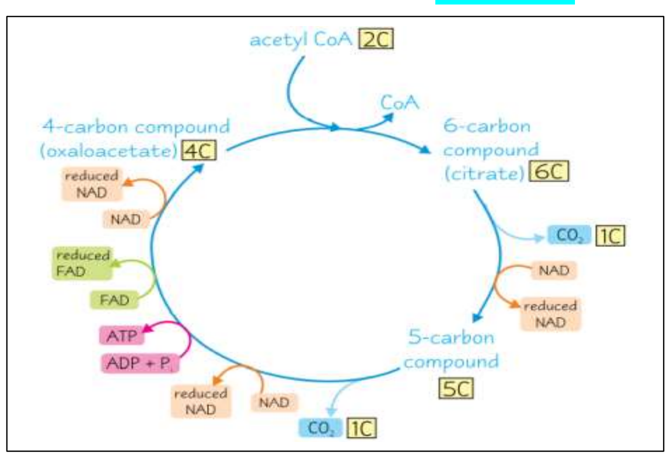
2- Linked reaction: "In mitochondrial matrix."

- Decarboxylation of 2x pyruvate (1 carbon atom are removed in the form of CO2) so 2x molecules of CO2 are produced.
    - ➢ Giving us 2x acetate.
- 2x NAD will be reduced by hydrogen from pyruvate giving us 2x NADH.
- 2x Acetate combines with coenzyme A to form 2x acetyl CoA.

3- Kreb's cycle: "In mitochondrial matrix."

- Acetyl CoA combines with oxaloacetate (4 carbon compound) to form citrate (6 carbon compound).
- Decarboxylation of citrate to release 1 CO2 and 1 NADH.
    - ➢ Citrate will be converted to 5 carbon compound.
- Decarboxylation of 5 carbon compound to release 1 CO2 and 1 NADH.
    - ➢ 5 carbon compound will be converted to oxaloacetate.
- FAD is reduced by electrons from 4 carbon compound to give 1 FADH.
- NADH will be produced.
- 2 ATP will be produced.
- Then regeneration of oxaloacetate for cycle to continue.

N.B: Multiply the products x2.



4- Oxidative phosphorylation: "In inner mitochondrial membrane."

- H atoms released from NADH and FADH.
  - H atoms splits into protons and electrons.
- Electron pass across ETC releasing energy.
- Energy is used to pump protons from mitochondrial matrix into the intermembrane space.
  - To create proton gradient.
- Hydrogen ions diffuse back into the matrix by chemiosmosis through ATP synthase.
  - Phosphorylation of ADP to form ATP.
- Oxygen act as final electron acceptor.
- Oxygen is reduced to water (oxygen, protons and electrons).

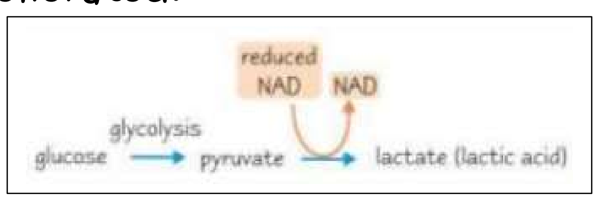
✓ Role of oxygen in aerobic respiration:

- Oxygen is needed in oxidative phosphorylation.
- Oxygen act as final electron acceptor.
- Reduced to water… for regeneration of NAD and FAD.

Anaerobic respiration:

- Lactate fermentation:

- Regeneration of NAD is achieved by pyruvate from glycolysis.
- Glycolysis produces 2 pyruvate, 2 NADH and 2 ATP.
- 2 pyruvates will be reduced to convert it to 2 lactate.
- Then oxidation of NADH to be regenerated.



✓ Animals break down lactate by:

1- Converting it to pyruvate.

2- liver cells convert lactate back to glucose.

- **Why anaerobic respiration produces less ATP?**
- Only glycolysis take place.
- Oxygen is absent so ETC won't take place.
- Reduced NAD is used to reduce pyruvate to lactate.

- **Oxygen debt:**
- After stopping exercise oxygen consumption increase.
- To breakdown lactate produced from anaerobic respiration.
- Lactate/lactic acid lowers the pH of blood.
- Breathing rate increases to transport lactate to the liver to be broken down.

- **Respiratory substrates:**

1- Glucose.

2- Lipids:

- Glycerol is phosphorylated and enter glycolytic pathway as GALP.
- Fatty acids hydrocarbon chains are broken down.
- Two carbons are broken at a time to produce acetyl CoA.
- Then it will enter the matrix to be used in Kreb's cycle.

3- Proteins:

- Deamination takes place in liver to remove amine group.
- Remaining organic part is converted into pyruvate or acetyl CoA (depending on R groups).
- Then enters Kreb's cycle.
- Used in case of starvation.

- **Respiratory substances:**

$$RQ = \frac{\text{Carbon dioxide produced}}{\text{Oxygen used}}$$

- RQ of glucose= 1
- RQ of triglyceride= 0.7
- RQ of proteins= 0.9

- RQ 1 or less than 1 is aerobic.
- RQ more than 1 is mixture.
- Very low RQ values are in photosynthetic organisms.

- **Tendons:**

- Attaches a muscle to a bone.
- No elastic fibers.
- So they don't stretch when the muscle contract.
- So the bone is moved.

- **Ligaments:**

- Attaches a bone to bone/holds bone together.
- Have elastic fibers.
- So bones can move at the joint without dislocation.
- Have collagen to provide strength.

- **Antagonistic pairs:**

- Muscles work in pairs called antagonistic pairs.
- They work opposite to each other, when one contract the other relax.
- As muscles pull but never push.

✓ **Why muscles occur in antagonistic pairs?**

- Muscles cannot extend themselves.
- Need opposing muscle to extend.
- Antagonistic muscle allows control of movement.

- **Muscle contraction:**

- Sarcoplasmic reticulum contains calcium ions.
- Calcium ions binds to troponin.
- Cause the movement of the tropomyosin.
- Which exposes the binding site of myosin head on the actin filament.
- Forming actomyosin cross bridges.
- Myosin head changes its shape....pulling the actin filament.
- ATP attaches to myosin head...breaking actomyosin cross bridges.
- Actin filament slide...causing the sarcomere to shorten.
- This is called sliding filament theory.

➢ There are two types of fibres:

Slow twitch fibre	Fast twitch fibre
More myoglobin	Less myoglobin
high capillary density	Less capillary density
Contract slowly, long period	Contract more rapidly, short period
More ATP (aerobic respiration)	Less ATP (anaerobic respiration)
Less glycogen	More glycogen

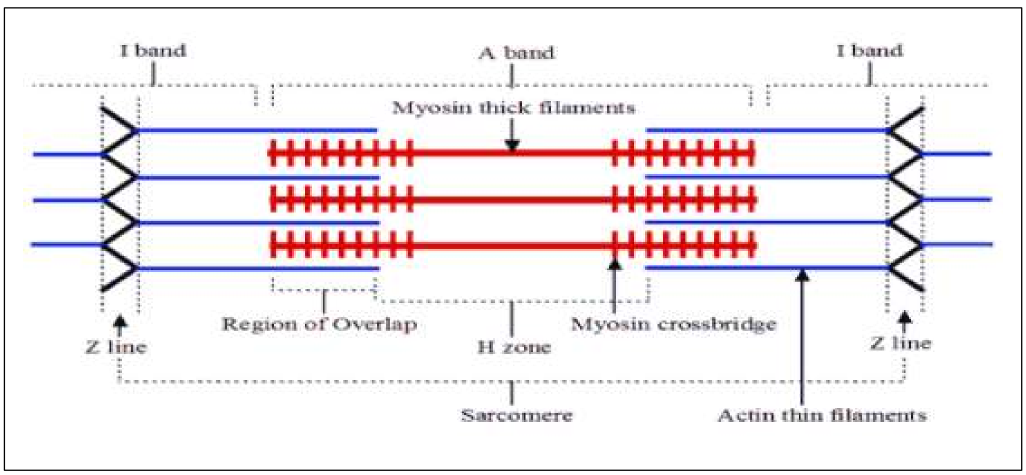
➢ Muscle structure:

Muscle fibres are made from:

- Myofibrils.
  - ➢ Each myofibril is made from sarcomeres.
- Sarcolemma (cell surface membrane), sarcoplasm (cytoplasm), mitochondria and sarcoplasmic reticulum.

➢ Sarcomere:

- Made up of actin (thin filament) and myosin (thick filament).
- I band= formed of actin only.
- A band= formed of actin and myosin.
- H zone= formed of myosin only.



Homeostasis:

- Maintenance of conditions inside the body near the constant level.
- By negative feedback mechanism.

Negative feedback mechanism:

- Mechanism that returns a change away from normal value to normal value.
- Keeping a constant value.

Control temperature by negative feedback mechanism:

- Change in temperature away from normal value act as a stimulus.
- Detected by thermoreceptors in skin/hypothalamus.
- Sending impulses to hypothalamus.
- Hypothalamus act as a thermoregulatory center.
- Hypothalamus send nerve impulses to effector.
- Temperature returns back to normal by homeostasis.

Homeostasis

In case of increase in temperature	In case of decrease in temperature
Vasodilation: - Arterioles in skin gets wider. - So more blood flow near the skin surface. - So more heat lost to surroundings.	Vasoconstriction: - Arterioles in skin gets narrower. - So less blood flow near the skin surface. - So less heat lost to surroundings.
Increasing sweat production: - So more heat loss by water evaporation from skin surface.	Decreasing sweat production: - So reduce heat loss by reducing water evaporation from skin surface.
Hair erector muscle relaxes: - So they lie flat reducing layer of insulation.	Hair erector muscle contract: - So trap layer of warm air.
Decreased adrenalin production: - Decrease in respiration so less heat generated.	Increased in adrenalin production: - Increase in respiration so more heat generated.
Inhibition of shivering: - So less heat generated.	Shivering involve muscle contraction. - So heat generated and absorbed by blood.

➢ Hormones:

Transcription factors:

- Control/involved in transcription of genes.
- Proteins acting as chemical messenger.
- TF binds to DNA at promoter.
- Activators increase rate of transcription and repressor decrease rate of transcription.

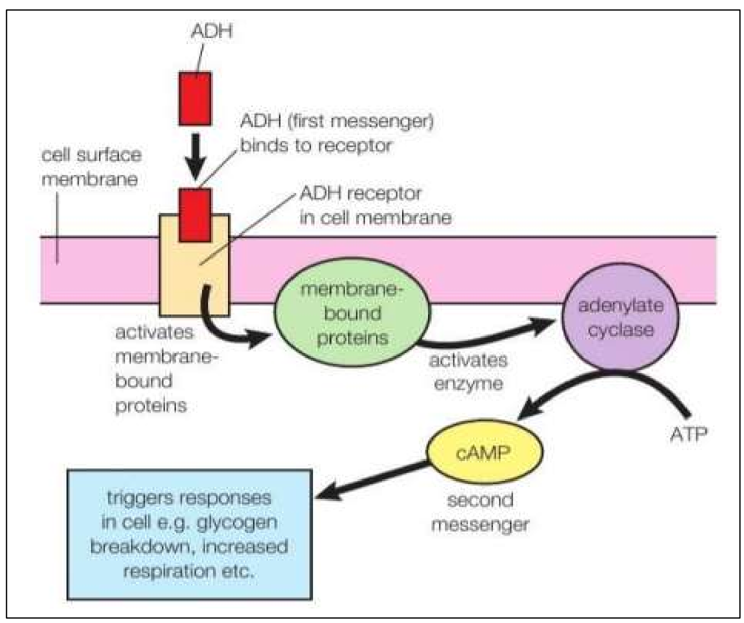
✓ There are two ways in which hormones can have their effect:

1- Release of second messenger.

2- Hormone that enter the cell.

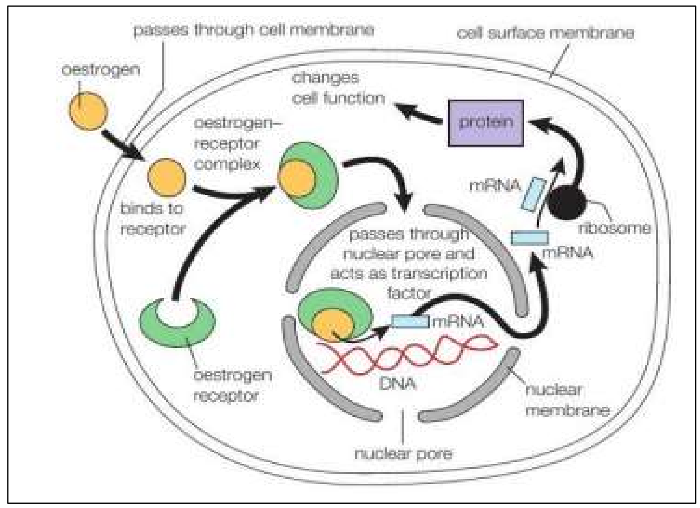
1- Peptide hormones (release of second messenger):

- They can't cross the cell membrane.
- They bind to specific receptors on the cell surface membrane causing a change in the shape of the receptor.
    - ➢ Resulting in the formation of a second messenger inside the cell.
- Second messenger activates number of different enzymes.
- Second messenger (Cyclic AMP) bind to other chemical which pass into the nucleus and act as DNA transcription factors.



2- Steroid hormones (hormones that enter the cell):

- They are lipid soluble so they pass through phospholipid bilayer.
- Then act as internal messenger itself.
- Hormone bind to receptors inside the cell and form hormone receptor complex and passes into the nucleus.
- Hormone receptor complex act as TF regulatory gene expression.



➢ Endocrine vs nervous system

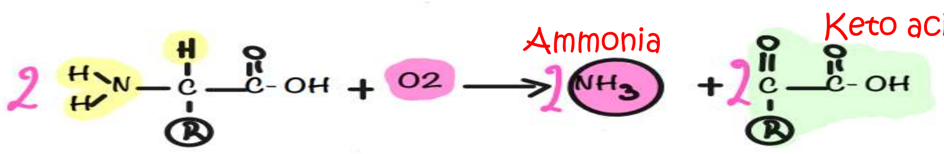
	Endocrine system	Nervous system
Similarities	Both involve chemicals	
	Both involve cell signalling	
	Both involve signal molecule binding receptor	
Differences		
Communication	By hormones	By nerve impulses
Nature of communication	Chemical	Electrical impulses
Mode of transmission	Blood	Neurones/nerve cell
Transmission speed	Slower	Faster
Duration	Long lasting	Short lived
Effects	Wide spread	Localised

Excretion:

It's the removal of toxic substance, waste products of metabolic reactions and the removal substances found in excess out of the body.

Deamination:

Break down of excess amino acids in liver, to remove amine group (NH_2) forming ammonia so ammonia combine with CO_2 in urea cycle forming urea.



Ammonia is highly toxic so converted into urea.

By adding CO_2 using ATP.

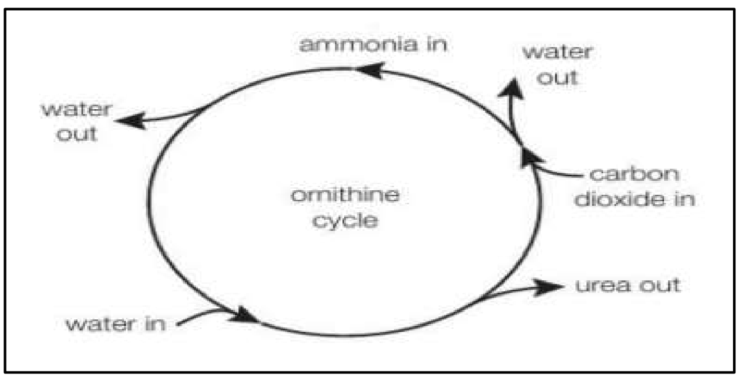
Keto acid is used in Kreb's cycle.

- Respired to produce ATP.
- Converted into glucose.

The ornithine cycle:

$2NH_3 + CO_2 ..... CO(NH_2)_2 + H_2O$

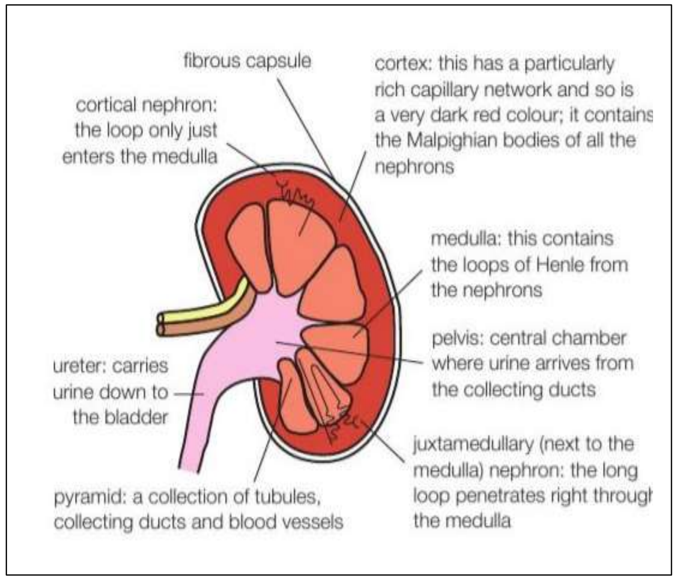
Urea



➢ Kidneys

Function of the kidney:

- Excretion, removal of urea from the body.
- Osmoregulation, control water potential of the blood that passes through them by ADH.
- ADH produced in hypothalamus, secreted from pituitary gland.
- To target collecting duct and DCT.



➢ Ultra-filtration:

- The blood comes in the afferent arteriole under high pressure.
- The efferent arteriole has a smaller diameter than afferent arteriole.
- Building up a pressure in the glomerulus forcing fluid out of the glomerulus through capillary pores into Bowman's capsule.
- Basement membrane act like a selective barrier to allow water and ions to pass through but large molecules don't pass like RBCs, WBCs, large plasma proteins.
- The wall of the capsule is made up of special cells called podocytes.
- The foot like extensions form gaps known as slit pores, molecules will pass through slit pores into Bowman's capsule.

➢ PCT:

1-Microvilli: increase surface area for absorption of sodium ions and glucose.

2-Mitochondria: to produce ATP for active transport of NA+ out of cells.

3-Tight junctions: formed between adjacent cells so that fluid can't pass between cells.

4-Folded basal membrane: with many transport proteins (sodium pump).

5- Carry co transporters.

- Selective reabsorption:

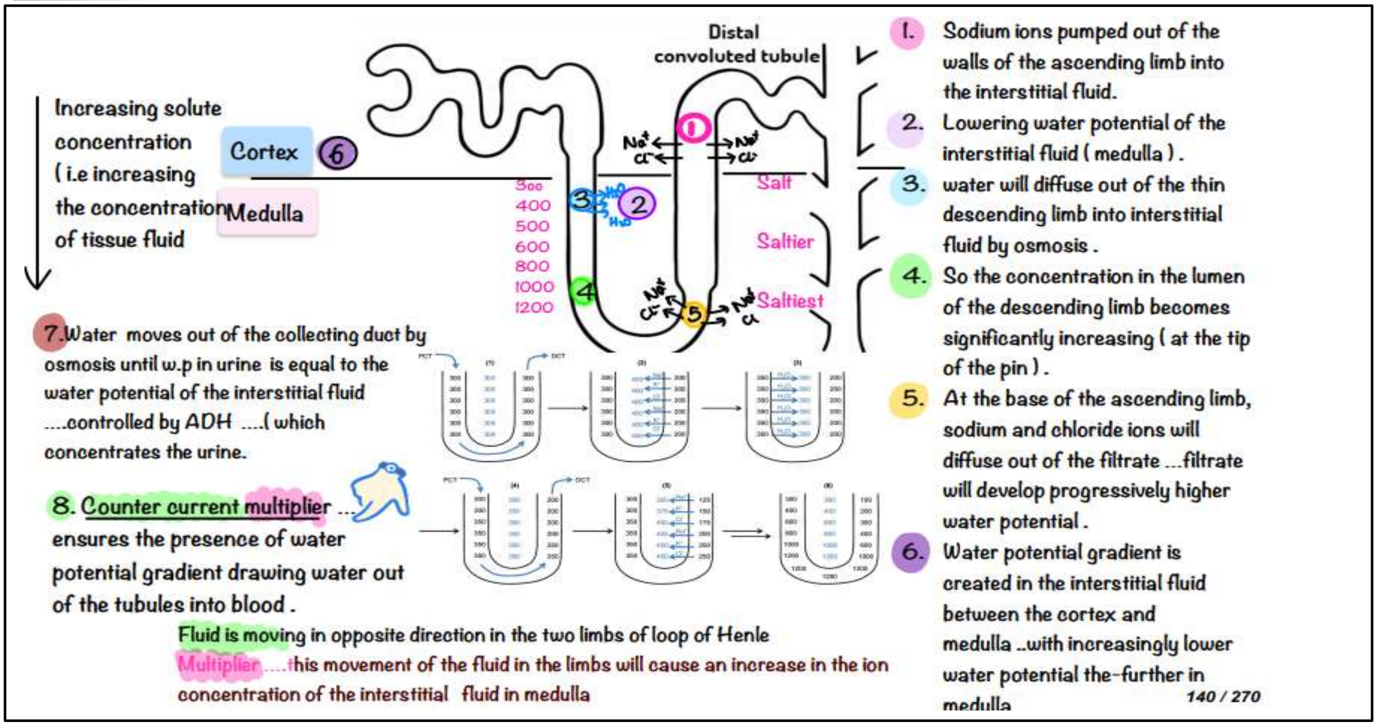
- After the filtrate has entered the nephron, the next step is to return most of what has been removed from the blood back.
- Sodium ions pumped out of the cells into blood against concentration gradient using ATP.
  - Thus, decreasing sodium ion concentration inside the cells.

- So sodium ions will diffuse into PCT cells through sodium co transport from the filtrate down concentration gradient.
- Glucose and amino acids enter PCT by secondary active transport.
- Glucose, amino acids, vitamins, and hormones are reabsorbed completely by active transport.
- Some sodium ions can also be reabsorbed by active transport.
- Chloride ions can also leave by diffusion.

- The co transport of solutes into the cytoplasm of epithelial cells increase concentration of the cytoplasm.
- So water moves by osmosis down water potential gradient from filtrate in the PCT lumen into the epithelial cell cytoplasm through the epithelial cell membrane, which in turn enters the blood by osmosis.

- Loop of Henle: "It has descending and ascending limbs."

- Loop of Henle act as a countercurrent multiplier to produce concentrated urine.
  - As the flow of filtrate in tubules is in opposite direction.
- Ascending limb, with thick walls that are impermeable to water and permeable to Na+.
- Na+ are actively pumped out of loop of Henle to tissue fluid/interstitial fluid.
- Descending limb, which is narrow with thin walls that are highly permeable to water and sodium ions.
- Water moves down water potential gradient by osmosis from loop of Henle to tissue fluid/ interstitial fluid.
- To increase the concentration gradient between tubular fluid and interstitial space.
- As filtrate moves down the descending limb it gets more concentrated.



➢ Distal convoluted tubule:

- It's permeable to water, but the permeability varies with the levels of antidiuretic hormone (ADH).
- If there is not enough salt in the body, sodium may be actively pumped out of the tubule with chloride ions.

➢ The collecting duct:

- Permeability to water varies according to ADH.
- Water moves out of the collecting duct with high level of sodium and chloride ions, and the urine becoming more concentrated.

✓ How ADH is involved in the control of water potential of blood?

- ADH is produced in hypothalamus and released from pituitary gland.
- Osmoreceptors (hypothalamus) detect change in water potential of blood.
- ADH is released and penetrate collecting duct.
- ADH act on DCT and CT (collecting duct) increasing the permeability to reabsorb water back to the blood.

➢ The urine:

- Collected in the pelvis then passes to the ureter then to be stored in the bladder.
- Volume of urine is variable according to what is taken in the body and activity level.
- Glucose and protein never appear in urine.

- Cardiac muscle is myogenic, it can contract and relax without any external stimulation.

1- Diastole (0.4s):

- Atria and ventricles are relaxed.
- Atrioventricular valve is open to allow filling of the heart.
- Semilunar valves closed.

2- Atrial systole (0.1s):

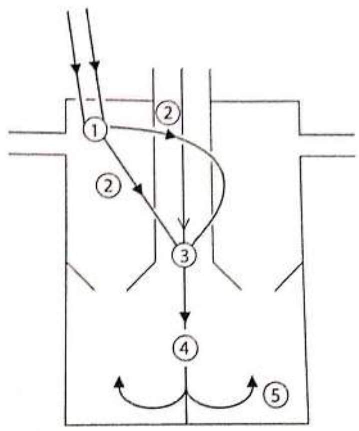
- Contraction of both atria and ventricles relaxes.
- Atrioventricular valve open.
- Semilunar valve closed.

3- Ventricle systole (0.3s):

- Contraction of both ventricles and atria relax.
- Ventricles contract.
- Atrioventricular valve closed.
- Semilunar valve open.

➢ Regulation of heartbeat:

- Process starts in the SAN, which is in the wall of the right atrium.
- Initiate depolarization.
- Acting as a pacemaker.
- Sending waves of excitation across the muscles in atrial walls.
- Stimulating both atria to contract.
- A band of non-conducting collagen tissue prevents the waves of electrical activity from being passed from the atria to the ventricles.
- Instead, waves of excitation are transferred from SAN to AVN.
- AVN found in the septum.
- Delay waves of excitation/depolarization.
- Receive waves of excitation and send them to bundle of His.
- Then purkyne tissue by time delay of 0.1s ensuring that atria contract before ventricles.
- Bundle of His and purkyne tissue spread waves of excitation down to the base of the septum.
- Both ventricles contract from base and upwards.



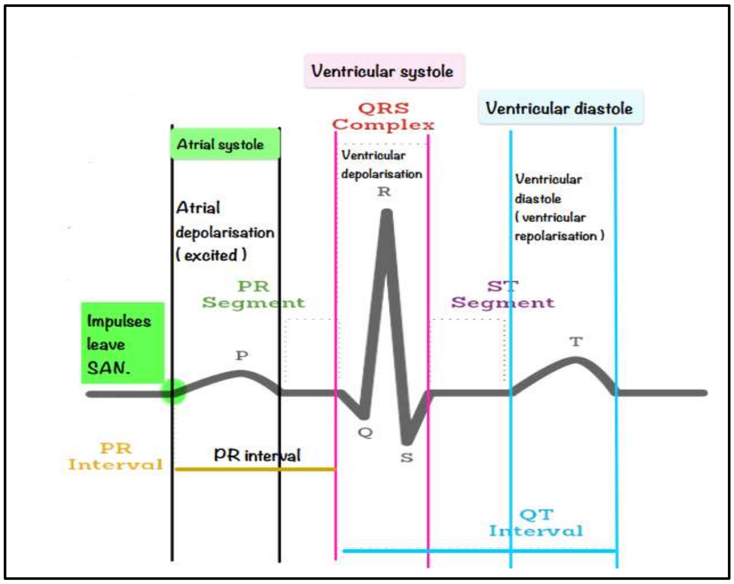
The diagram shows different stages in the passage of electrical activity through the heart.

The arrows and numbers represent the different stages.

- ✓ The number in the diagram that represents the bundle of His is: 4
- ✓ The number in the diagram that represents where the delay occurs is: 3

> ➢ Electrocardiograph (ECG):

- It shows heart rate/waves.
- As these are the waves of electrical activity in the heart.
- Over a period of time or cardiac cycle.



> ➢ Cardiovascular control centre receives impulses from:

1- Chemoreceptors:

- Higher centre: Medulla oblongata.
- Found in: Aortic body/ carotid body/ medulla.
- It's sensitive to levels of CO_2 in blood and pH.
- Function: maintain normal blood gases levels (CO_2).

2- Baroreceptors:

- Higher centre: Medulla oblongata.
- Found in: Carotid artery in neck and aorta.
- It's sensitive to change in blood pressure.
- Function: maintain normal blood pressure.

➢ Parameters measuring the Cardiac Output:

- Stroke Volume = volume of blood pumped by the heart in 1 beat.
- Heart Rate = number of heart beats per minute.
- Cardiac Output = volume of blood pumped by the heart in 1 minute.
- Cardiac output = stroke volume x heart rate

➢ Adjusting the Cardiac Output:

A- Nervous control of the heart

- It is achieved by the cardiovascular control centre which is found in the medulla oblongata.
- This centre is connected to the autonomic nervous system which has 2 arms:

  1- Sympathetic nervous system: carries excitatory impulses causing more stimulation of the SAN.

  - Increasing heart rate and cardiac output.

  2- Parasympathetic nervous system: carries inhibitory impulses to the SAN by releasing Acetyl choline.

  - Heart rate decrease.

B- Hormonal control of the Heart

- Adrenaline acts as a neurotransmitter on the SAN, increasing the frequency of excitation and speeding up the heart rate.

✓ Describe the changes in heart that causes an increase in cardiac output:

- Increase in heart rate.
- Increase in stroke volume.
- Increase in the activity of SAN.
- The time delay from AVN decreases.

Ventilation

Inspiration:

- The respiratory centre sends rhythmic impulses so diaphragm and external intercostal muscles contract.

Expiration:

Stretch receptors send the following impulses:

- Inhibitory impulses to inspiratory centre causing relaxation of diaphragm and external intercostal muscles.
- Excitatory impulses to the expiratory centre causing contraction of the internal intercostal muscles.

The respiratory centre receives impulses from:

1- Chemoreceptors:

- Higher centre: Medulla oblongata.
- Found in: Aortic body/ carotid body/ medulla.
- It's sensitive to levels of CO_2 in blood and pH.
- Function: maintain normal blood gases levels.

2- Baroreceptors:

- Higher centre: Medulla oblongata.
- Found in: Carotid artery in neck and aorta.
- It's sensitive to change in blood pressure.
- Function: maintain normal blood pressure.

➢ Chemoreceptors in case of increase in heart rate:

- Sensitive to levels of CO_2.
- As CO_2 concentration increase blood pH decrease and chemoreceptors detect the decrease in pH.
- So send impulses to CV center in medulla oblongata.
- To send more frequent nerve impulses down sympathetic nerve to heart (SAN) by releasing nor epinephrine.
- So SAN send more waves of excitation increasing heart rate/ increase blood flow to lungs so more CO_2 is removed.

➢ Baroreceptors after exercise:

- Heart rate is still high… increasing blood pressure.
- Causing baroreceptors to stretch again and send nerve impulses to CV centre in medulla oblongata.
- Sends more impulses across the parasympathetic nerves. Slowing down heart rate and vasodilation decreasing blood pressure again.

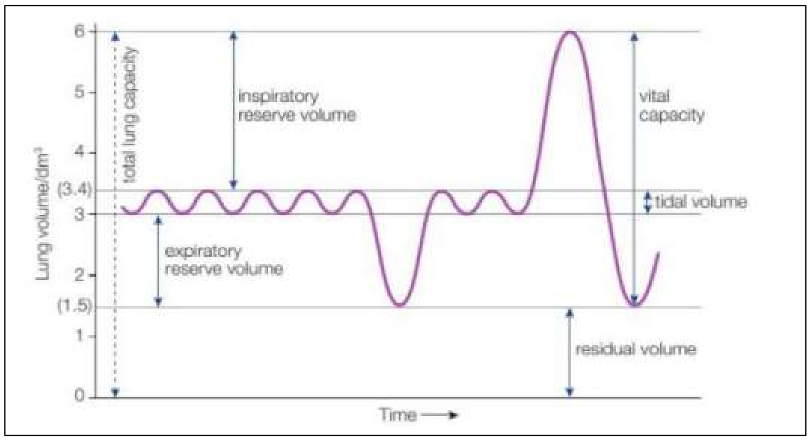
➢ Baroreceptors during exercise:

- Blood vessels are already dilated.
- Decreasing blood pressure.
- Reducing stretching on baroreceptors.
- Reducing stimulation from these receptors to the CV center (cardiovascular center).
- CV center send nerve impulses across sympathetic nerves to the (SAN) in the heart through nor epinephrine.
- Increasing heart rate and blood pressure (by constriction).

➢ Components of Lung Volume:

The components of lung volume are measured by the Spirometer.

VC = VT + IRV + ERV
TLC = VC + RV
IC = VT + IRV
Ventilation centre: Tidal volume × No. of breaths per minute)



➢ Definitions:

Tidal volume (VT): the volume of air that enters and leaves the lung in normal one breath.

Inspiratory reserve volume (IRV): the volume of air that you can take in above the normal inspired tidal volume.

Expiratory reserve volume (ERV): the volume of air that you can force out above the normal expired tidal volume.

Residual volume (RV): the volume of air remaining in the lungs after you have forcefully exhaled.

Vital capacity (VC): the maximum volume of air which you can breathe out by breathing out as hard as you can and then breathing in as deeply as possible.

Total lung capacity: the sum of the vital capacity and the residual volume.

Inspiratory capacity: the volume of air that can be inspired from the end of normal expiration.

Ventilation rate (VE): Volume of air breathed in a minute.